

Web Scraping

UNIT – I: Introduction to Web Scraping

1. What is Web Scraping? Explain its applications.
2. Compare Web Scraping vs Web Crawling.
3. What are the legal and ethical considerations in web scraping?
4. Discuss the challenges faced during web scraping.
5. Explain the structure of a typical web page (HTML, DOM).

UNIT – II: Tools and Libraries for Web Scraping

1. Explain how to install and set up Python environment for scraping.
2. What is BeautifulSoup? Explain its basic usage with example.
3. Explain the role of requests library in web scraping.
4. Compare different web scraping libraries: BeautifulSoup vs Scrapy vs Selenium.
5. How do you handle HTTP errors and status codes?

UNIT – III: Scraping Techniques

1. How to extract text, links, and images using BeautifulSoup?
2. Write Python code to scrape data from a table on a webpage.
3. How to handle nested HTML tags and classes?
4. Explain how to use CSS selectors and XPath in scraping.
5. How to save scraped data into CSV or JSON format?

UNIT – IV: Dynamic Websites and Selenium*

1. Why is Selenium used in web scraping?
2. Write code using Selenium to fill a form and submit.
3. Explain how to interact with dropdowns, checkboxes, and buttons using Selenium.
4. How to wait for elements to load dynamically using Selenium?
5. Compare static scraping (requests + BeautifulSoup) vs dynamic scraping (Selenium).

UNIT – V: Advanced Topics & Best Practices

1. How to bypass restrictions like CAPTCHA or bot detection?
2. What are proxies and user agents? How are they used in scraping?
3. Explain how to use regular expressions (regex) in scraping.
4. What are APIs? When should you prefer API over scraping?
5. Best practices to write efficient and polite web scrapers.